

VASTCalc TUI Playbook

An end-to-end sizing walkthrough, from customer requirement to a laid-out rack

Scenario

A customer requires VAST NCP storage for a GPU cluster running the **DGX-C** workload profile on **NVIDIA GB200/GB300** platforms. They specify **12D EBox V2 TLC** hardware at **30TB** SSD density, sized for an initial **3-rack** deployment. This playbook walks through sizing that requirement with `vastcalc_tui` (the ncurses frontend to VASTCalc) and then laying the resulting EBoxes out in a physical rack elevation — a single, complete, start-to-finish solutioning example.

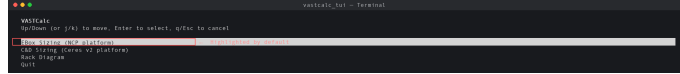
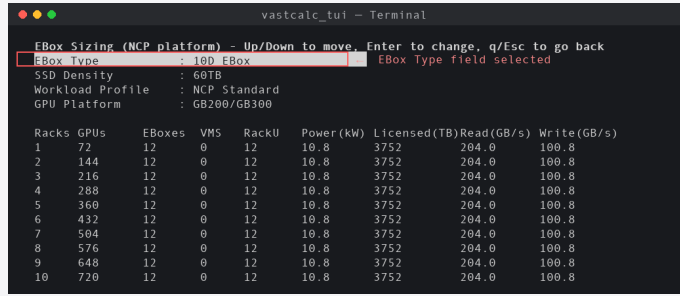
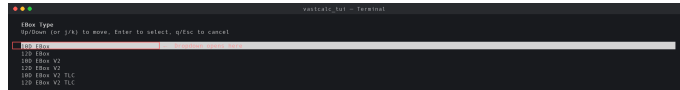
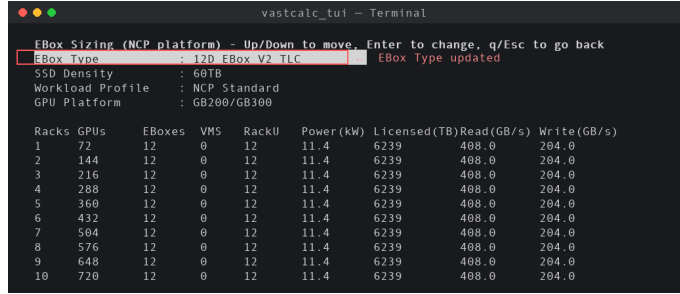
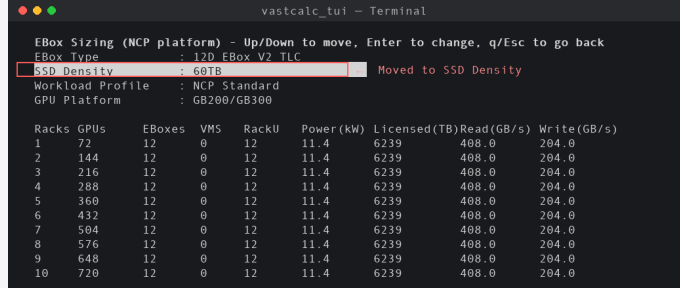
Result

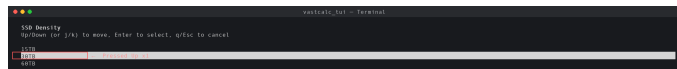
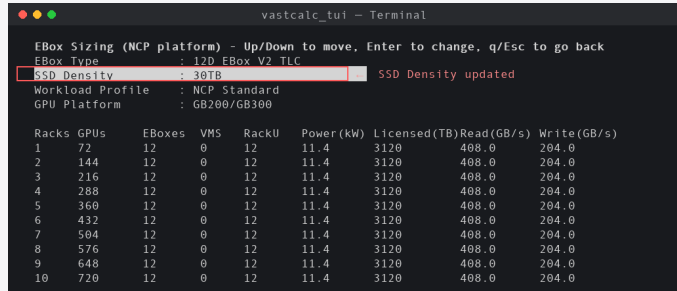
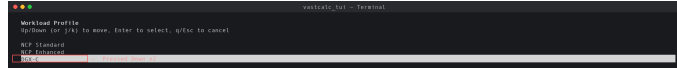
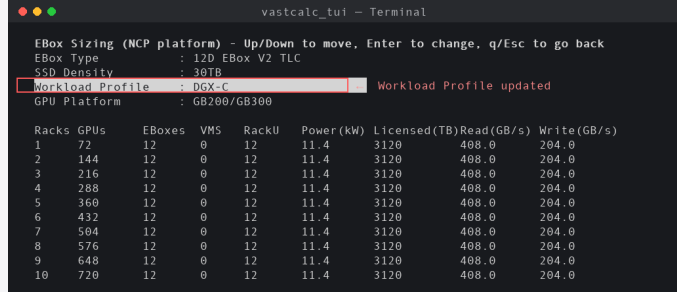
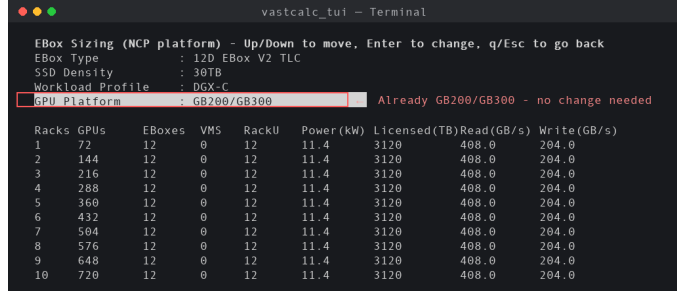
216 GPUs across 3 racks require **12 EBoxes** (the sizing floor at this scale), drawing **11.4 kW** and delivering **3120 TB** of licensed usable capacity at 408.0/204.0 GB/s read/write. The Rack Diagram screen confirms this independently: 12 physical 12D-EBOX-V2-TLC-30 units placed in one 42U rack report the same 11.4 kW total.

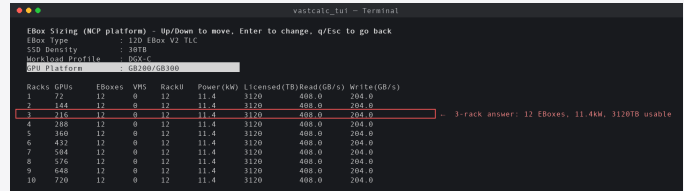
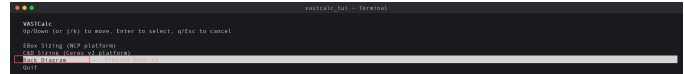
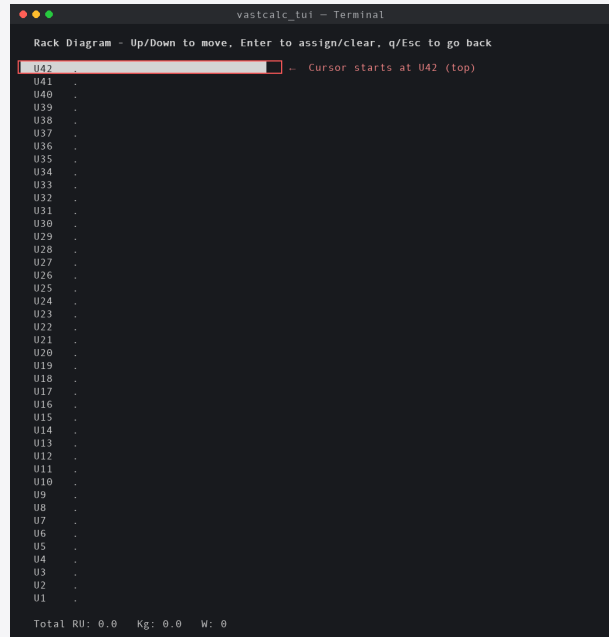
How to read this playbook

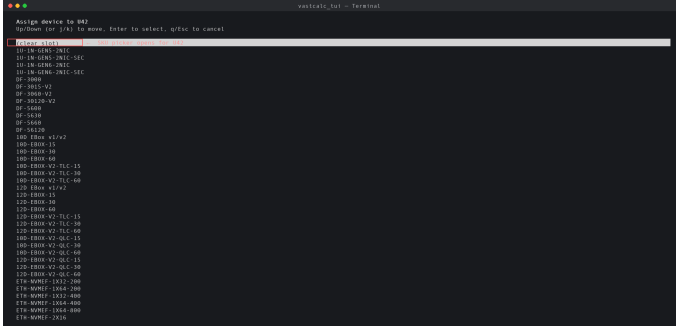
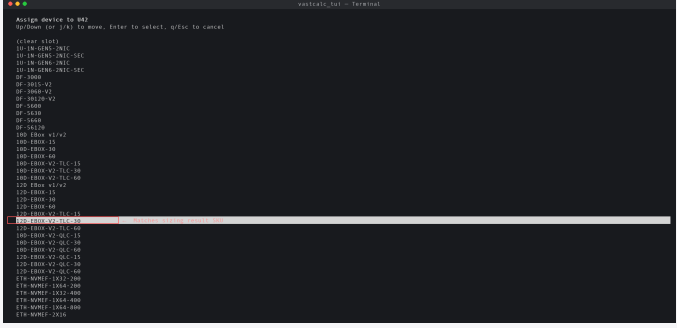
Each row below is one interaction with the TUI. The **Input Sequence** column is the exact keystrokes sent; the **Screenshot** is the terminal's actual state immediately after, captured from a live run (via `tmux` driving a real `pty`) — not a mockup. A red box highlights the control being acted on in that step, with a callout naming it. Screens were captured at reduced terminal width for this document; the sizing tables and rack view render identically at any terminal size.

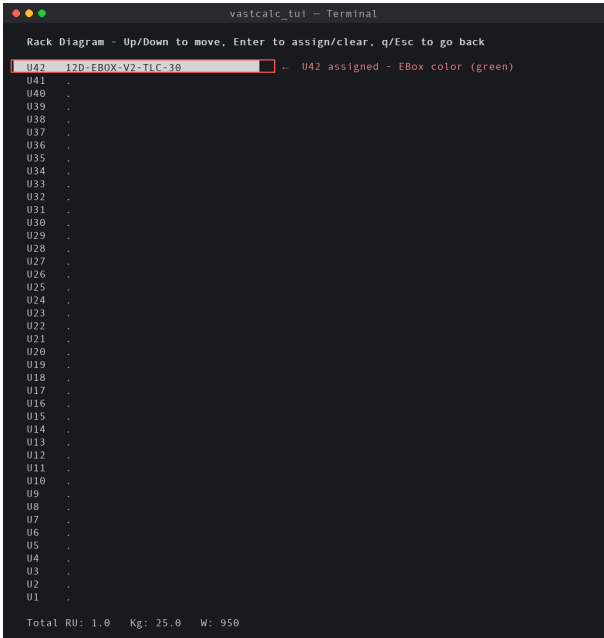
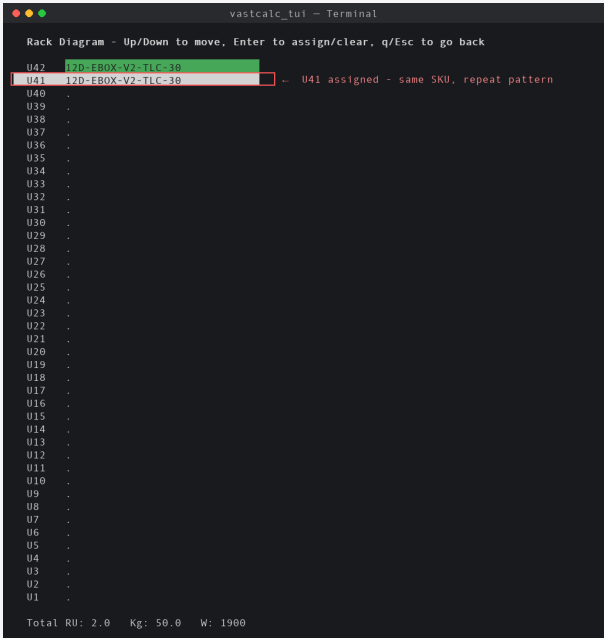
The C&D Sizing screen (for the Ceres v2 CBox/DBox platform) follows an identical four-field, list-menu interaction pattern and isn't separately walked through here.

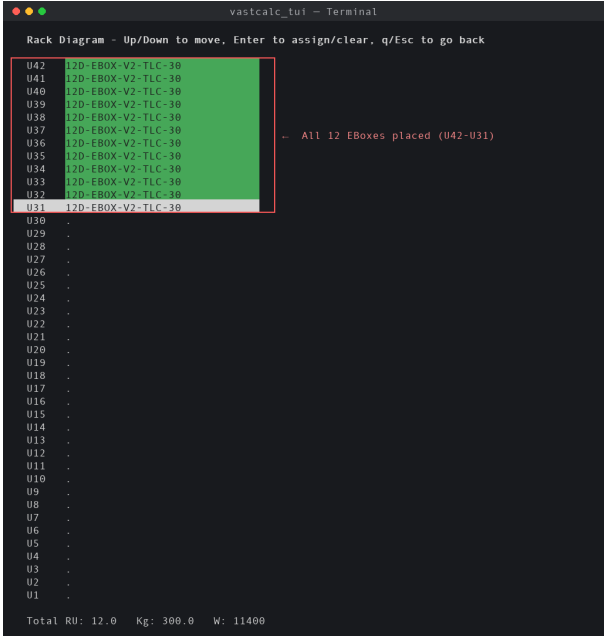
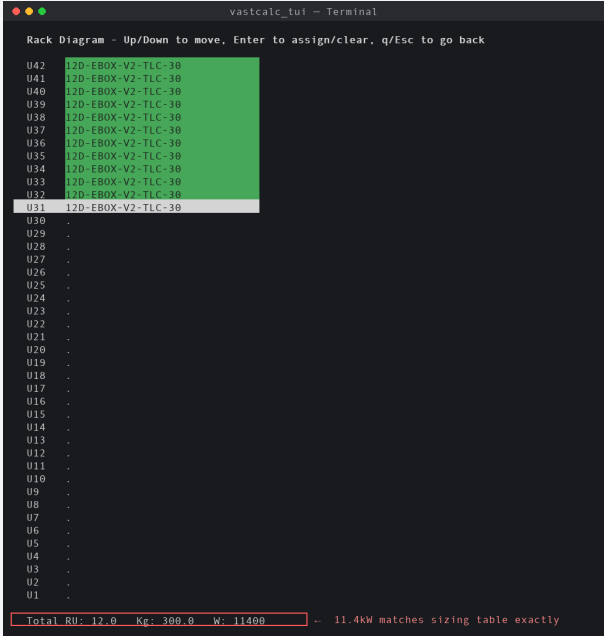
1	Launch the TUI. The main menu appears with EBox Sizing (NCP platform) already highlighted (it's the first item).	\$./vastcalc_tui	
2	Confirm the selection to enter the EBox Sizing screen. It opens showing the sheet's own defaults (10D EBox / 60TB / NCP Standard / GB200-GB300) and a 10-row sizing table that already reflects them.	Enter	
3	Open the EBox Type dropdown to change hardware to what the customer specified: 12D EBox V2 TLC.	Enter	
4	Navigate down to 12D EBox V2 TLC in the list.	Down x5	
5	Confirm the choice. EBox Type updates immediately and the whole sizing table recomputes live — Licensed capacity jumps to 6239 TB at the still-default 60TB density.	Enter	
6	Move the cursor down to the SSD Density field.	Down	

7	Open its dropdown and move up to 30TB — the density the customer specified.	Enter, Up	
8	Confirm. SSD Density updates to 30TB and Licensed capacity recomputes to 3120 TB.	Enter	
9	Move to Workload Profile , open its dropdown, and navigate to DGX-C per the requirement.	Down, Enter, Down x2	
10	Confirm. Workload Profile updates to DGX-C.	Enter	
11	Move to GPU Platform and check it: it's already GB200/GB300, which matches the requirement exactly — no change needed here.	Down	

12	Read the answer. All four inputs now match the customer requirement. The row for a 3-rack deployment reads: 216 GPUs, 12 EBoxes, 0 VMS nodes, 11.4 kW, 3120 TB licensed usable capacity, 408.0/204.0 GB/s read/write.	(read table – no input)	
13	Exit the sizing screen back to the main menu, then navigate down to Rack Diagram to lay out the physical result.	q, Down x2	
14	Enter the Rack Diagram screen. An empty 42U rack is shown, cursor starting at the top slot, U42.	Enter	

15	Open the device picker for U42.	Enter	
16	Navigate to the exact SKU the sizing answer requires: 12D-EB0X-V2-TLC-30.	Down x25	

17	Confirm. U42 is now assigned — shown here still under the cursor's reverse-video highlight, which is why its true EBox-green fill isn't visible yet.	Enter	
18	Repeat the identical pattern for U41: move down, open the picker, select the same SKU, confirm. With the cursor off U42 now, its green EBox color is visible — this is the repeatable unit of work for the rest of the rack.	Down , Enter , Down x25 , Enter	

19	Repeat that same four-keystroke pattern 10 more times (U40 down to U31) to place all 12 EBoxes the sizing answer called for.	(Down, Enter, Down x25, Enter) x10	
20	Verify end-to-end consistency. The rack's own totals footer reads 11.4 kW — the same number the sizing table produced in step 12, independently, from a completely different code path (RackLayout::totals() vs. EBoxSizingEngine). The two engines agree.	(read totals – no input)	

21	Exit the rack screen back to the main menu and select Quit .	q, Down x3	
22	Confirm. The process exits cleanly and the terminal is restored to normal — no leftover raw-mode state, no crash.	Enter	(terminal closes — tmux pane exits)